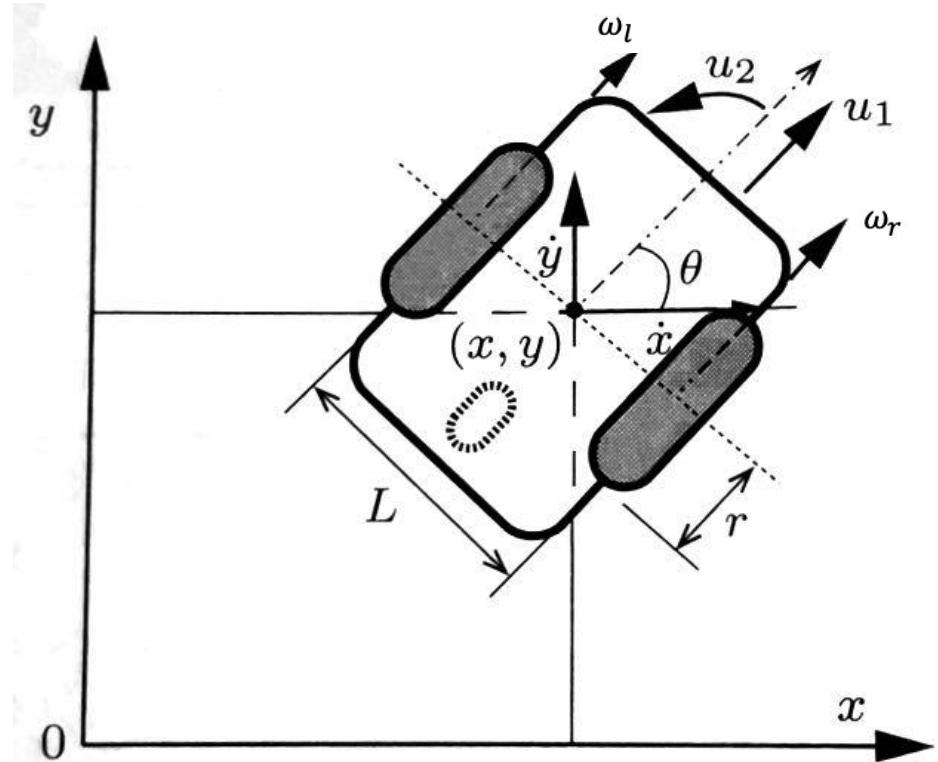

Mobile Robot 1

— Electric, Electronic and System
Engineering —

2-Wheel Drive Robot

2-Wheel Drive Mobile Robot

(x,y)	:robot position [m]
θ	:robot orientation [rad]
r	:radius of wheel [m]
L	:tread width [m]
ω_r, ω_l	:wheel speed [rad/s]
u_1	:translational speed [m/s]
u_2	:rotational speed [rad/s]

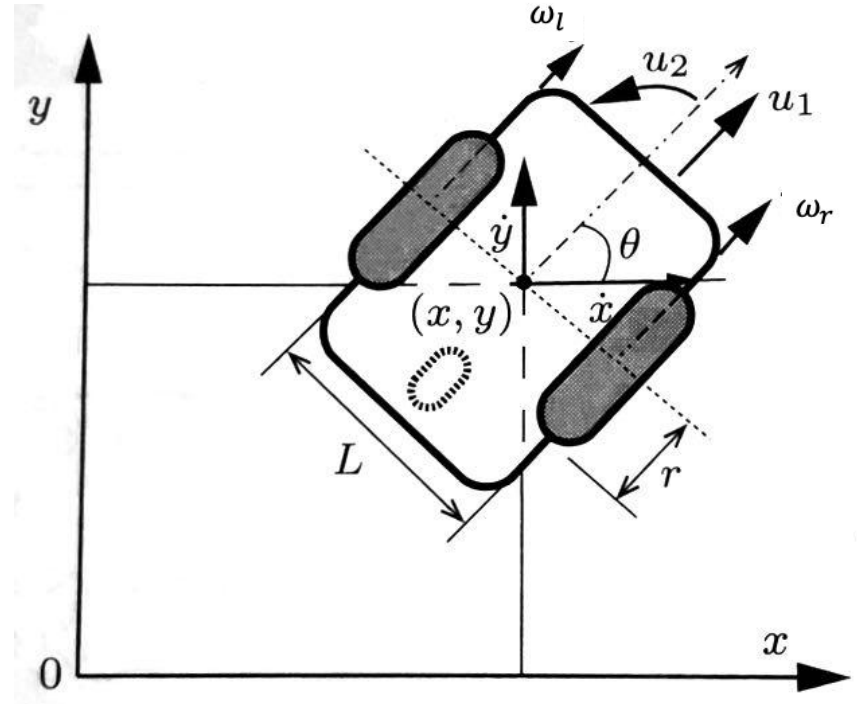


Introduce Translation Matrix

$$\mathbf{x} \equiv \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}, \quad \mathbf{u} \equiv \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\mathbf{A} \equiv \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix} \quad \mathbf{x} = \mathbf{A}\mathbf{u}$$

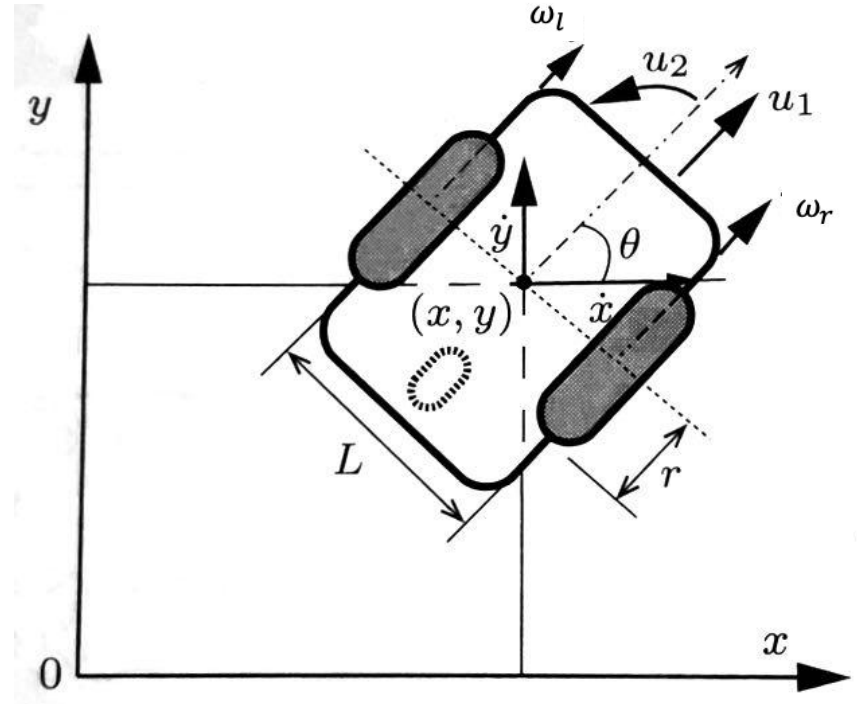


Translation from wheel speed to robot speed

$$r \frac{\omega_r + \omega_l}{2} = u_1 \quad \frac{r(\omega_r - \omega_l)}{L} = u_2$$

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} ? \\ ? \end{bmatrix} \begin{bmatrix} \omega_r \\ \omega_l \end{bmatrix}$$

$$B \equiv \begin{bmatrix} ? \\ ? \end{bmatrix} \quad \boldsymbol{\omega} \equiv \begin{bmatrix} \omega_r \\ \omega_l \end{bmatrix}$$



Robot speed in world coordinate

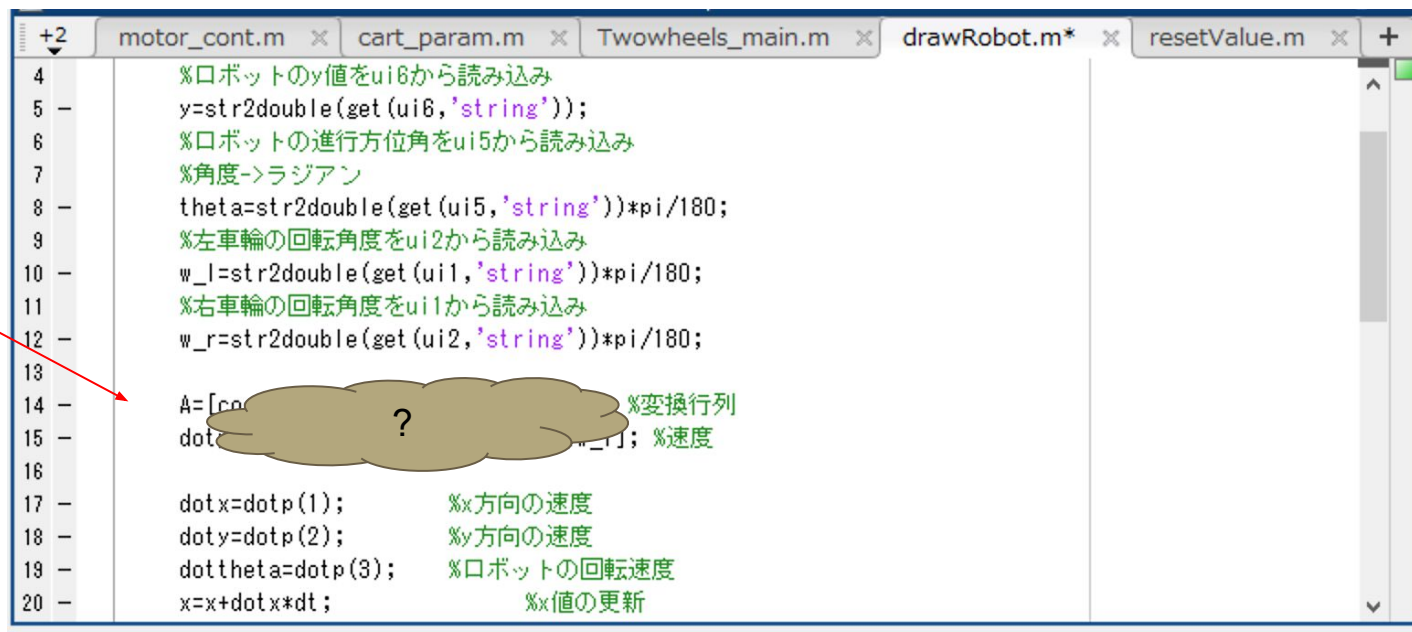
$$\mathbf{x} = A\mathbf{u} = AB\boldsymbol{\omega}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \frac{r}{2}\cos\theta & \frac{r}{2}\cos\theta \\ \frac{r}{2}\sin\theta & \frac{r}{2}\sin\theta \\ r/L & -r/L \end{bmatrix} \begin{bmatrix} \omega_r \\ \omega_l \end{bmatrix}$$

Robot speed in world coordinate was described by wheel speed

Simulate Robot Movement using m.file

In
drawRobot.m,
robot speed is
calculated in
this section

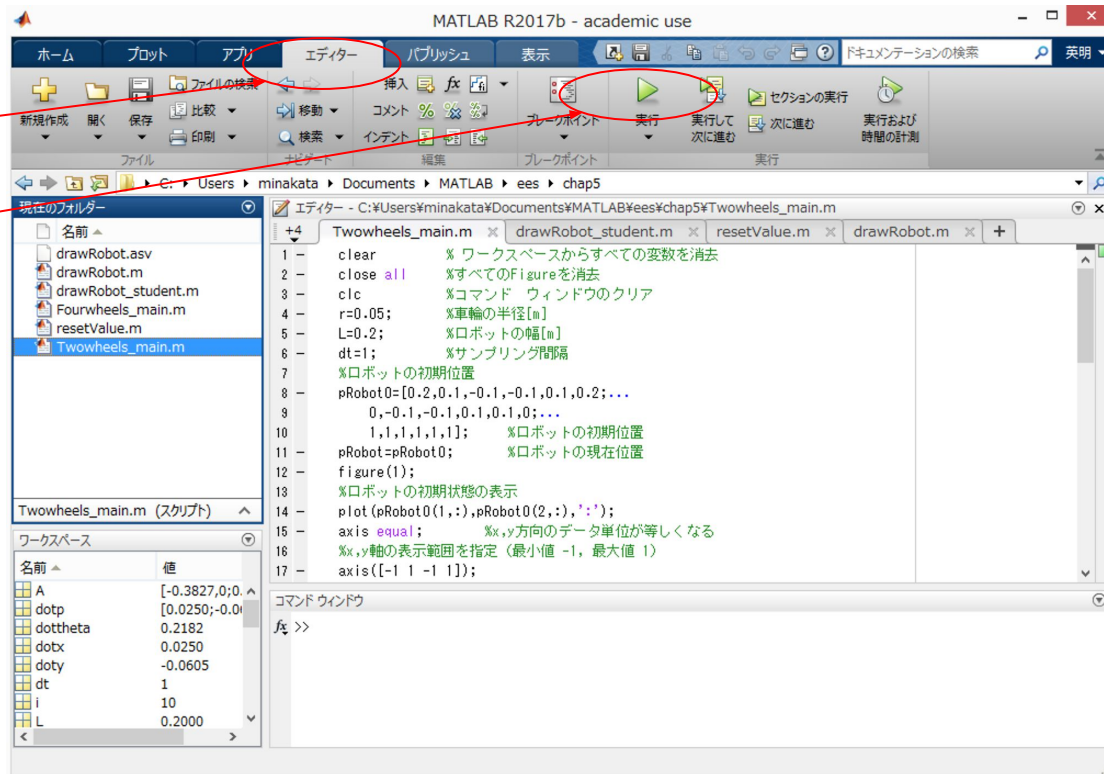


```
4      %ロボットのy値をui6から読み込み
5      y=str2double(get(ui6,'string'));
6      %ロボットの進行方位角をui5から読み込み
7      %角度->ラジアン
8      theta=str2double(get(ui5,'string'))*pi/180;
9      %左車輪の回転角度をui2から読み込み
10     w_l=str2double(get(ui1,'string'))*pi/180;
11     %右車輪の回転角度をui1から読み込み
12     w_r=str2double(get(ui2,'string'))*pi/180;
13
14     A=[cos(theta)  -sin(theta)  0; %変換行列
15       sin(theta)   cos(theta)  0; %速度
16       0            0          1];
17     dotx=dotp(1); %x方向の速度
18     doty=dotp(2); %y方向の速度
19     dottheta=dotp(3); %ロボットの回転速度
20     x=x+dotx*dt; %x値の更新
```

Start Drawing using m.file

choose
Twowheels_main.m

& Run

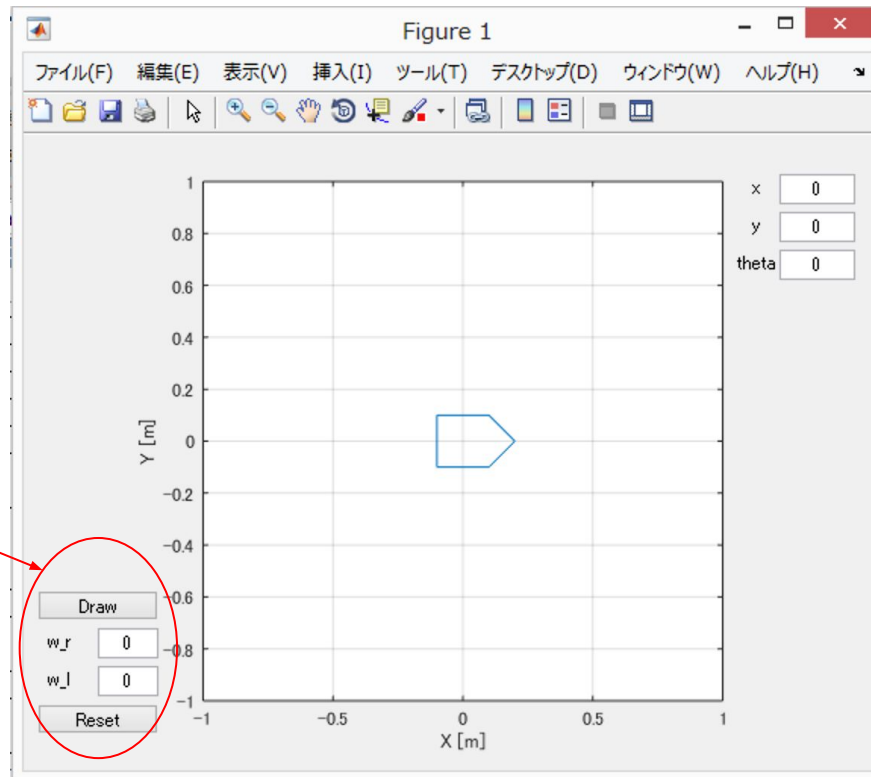


Drawing of the 2 Wheel Robot

Initial State:

wheel speed & robot position
are 0.

1. Input wheel speed in the
left bottom box.
2. Push [Draw]



Example

w_r : 50

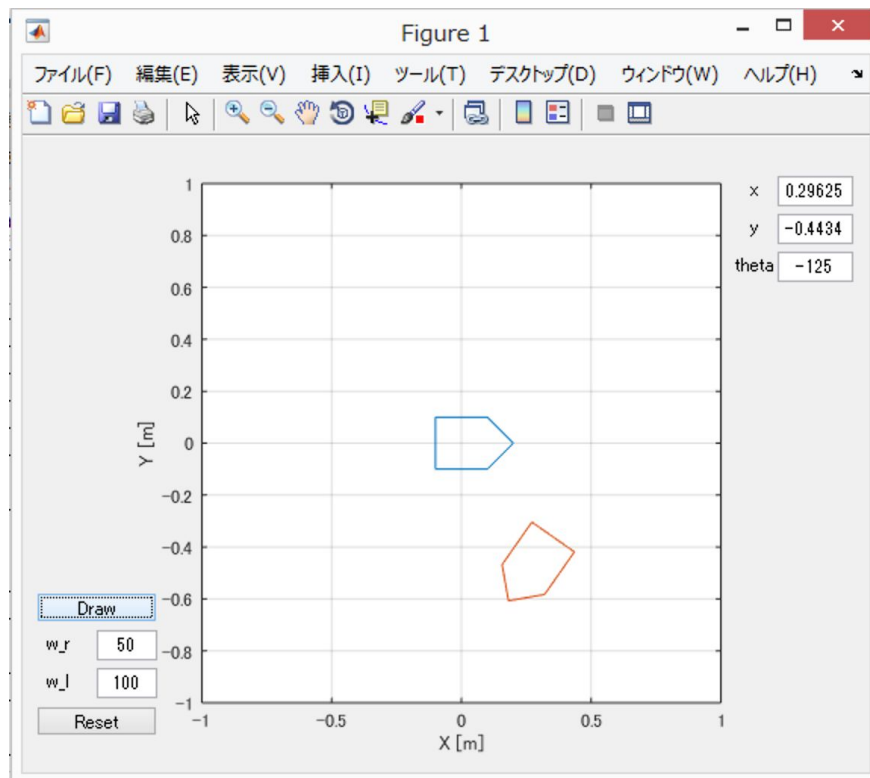
w_l : 100

→ moves forward & turn to right

Red Robot is the state **after 10s.**

You can continue the drawing with same (or another) wheel speed.

Push Reset to Initialize the robot.



Customize the simulation

r: 5cm
L: 20cm
dt: 1s (sampling time)

drawrobot.m draws 10
times using **for-loop**.

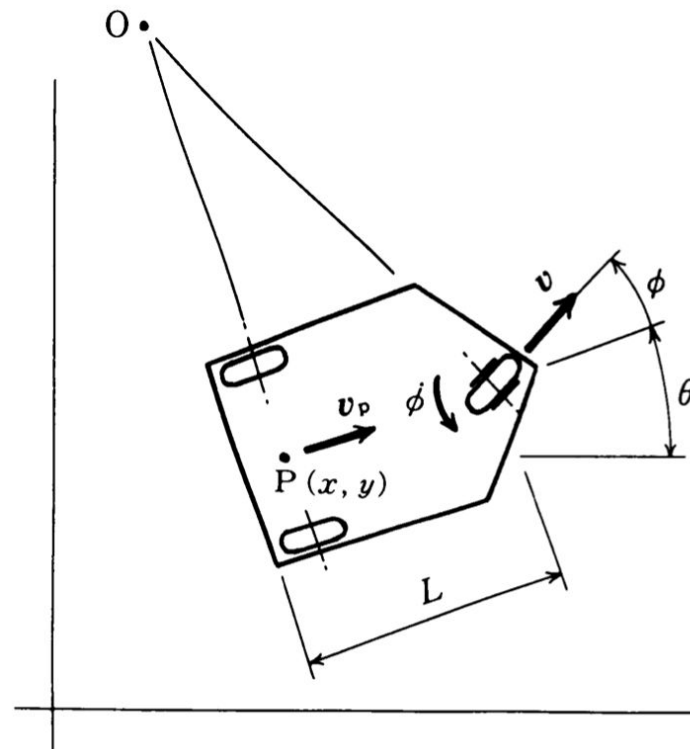
```
+4 Twowheels_main.m x drawRobot_studen
1 - clear % ワークスペースからすべての
2 - close all %すべてのFigureを消去
3 - clc %コマンド ウィンドウのクリア
4 - r=0.05; %車輪の半径[m]
5 - L=0.2; %ロボットの幅[m]
6 - dt=1; %サンプリング間隔
7 %ロボットの初期位置
```

```
+4 Twowheels_main.m x drawRobot_student.m x resetValue.m x drawRobot.m x
1 - for i=1:1:10
2 - %ロボットのx値をui7から読み込み
3 - x=str2double(get(ui7,'string'));
```

3 Wheel Robot (1S2D)

1 Steering 2 Wheel Drive Robot

(x,y)	:robot position [m]
(x_1,y_1)	:front wheel position [m]
φ	:steering angle [rad]
θ	:robot orientation [rad]
L	:wheel base [m]
v_p	:robot speed [m/s]
v	:front wheel speed [m/s]
u_1	:translational speed [m/s]
u_2	:steering speed [rad/s]



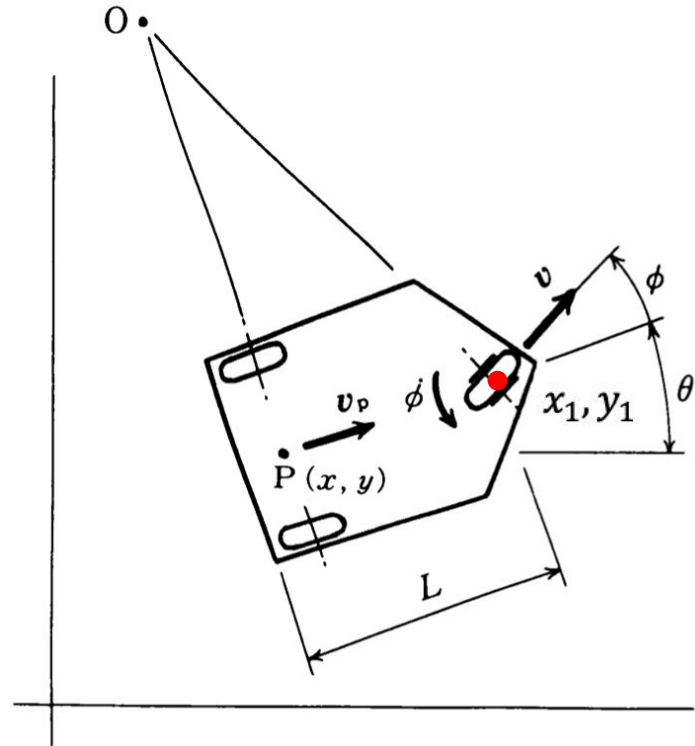
Constraints

$$\dot{x}\sin\theta - \dot{y}\cos\theta = 0$$

$$\dot{x}_1\sin(\theta + \phi) - \dot{y}_1\cos(\theta + \phi) = 0$$

$$x_1 = x + L\cos\theta, \quad y_1 = y + L\sin\theta$$

$$\dot{\theta} = \frac{u_1}{L}\tan\phi$$

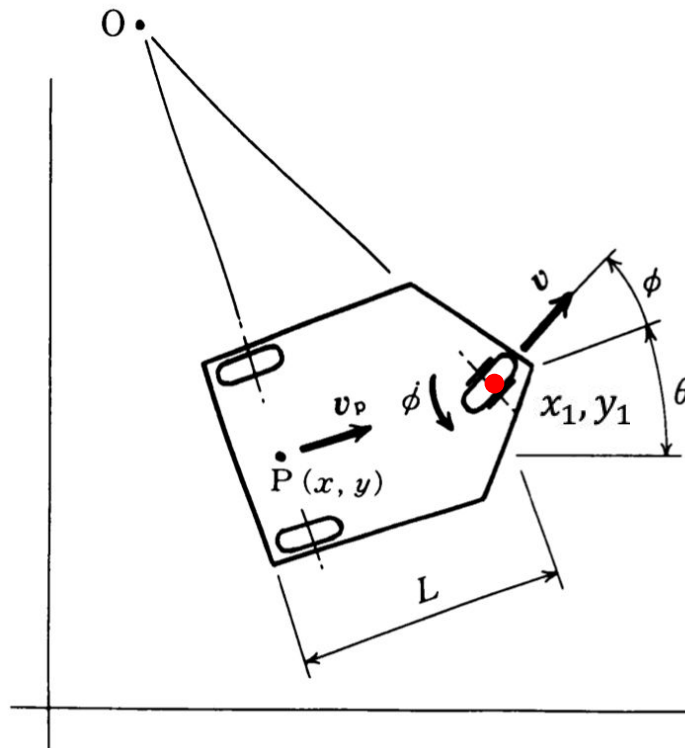


Introduce Translation Matrix

Input: u_1, u_2

State: x', y', θ', ϕ'

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ \frac{L}{0} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$



Next Topic

Robot will follow a trajectory using

- velocity input :u1
- steering angle speed input :u2

